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Physical activity profiles of attention-deficit/hyperactivity disorder symptoms among preschool children

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Abstract

Background Physical activity (PA) has gained attention as a potential behavioral marker for ADHD, yet findings remain inconsistent. Most research focuses on school-aged children, leaving a critical gap in understanding PA patterns in preschoolers with ADHD symptoms.

Aim The study aims to investigate the association of physical activity level and ADHD symptoms and focus on characterizing physical activity profiles of preschool children with ADHD symptoms.

Methods Totally 18 children with ADHD symptoms and 66 typically developing children were enrolled in this study. Physical activity was objectively assessed using actigraphy over five school days. Two-sample Wilcoxon rank-sum test was used to assess differences between groups. Multivariate regression models were further employed to control potential confounding variables.

Results Preschool children with ADHD symptoms had less time in most of the daily physical activities. Especially, ADHD groups had lower light (24.11 min *vs.* 27.25 min, $P=0.023$), moderate (16.56 min *vs.* 20.43 min, $P=0.009$) and moderate-to-vigorous physical activities (MVPA, 23.26 min *vs.* 29.20 min, $P=0.018$) than control group. Potential interactive effects among light, moderate, MVPA and routine afterschool physical activities were further investigated using multivariate regression analysis.

Conclusions The observed pattern of reduced PA in preschool children with ADHD symptoms may reflect behavioral characteristics associated with early ADHD manifestations. However, further longitudinal research is needed to determine whether activity differences could serve as potential early markers for ADHD.

Keywords Physical activity, Attention-deficit/hyperactivity disorder symptoms, Preschool children, Kindergarten settings, Extracurricular physical exercise

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Introduction

Attention-deficit/hyperactivity disorder (ADHD), a clinically significant neurodevelopmental condition characterized by persistent inattention, hyperactivity, and impulsivity [1], affects approximately 7.6% (95% CI: 6.1–9.4) of children aged 3–12 years and 5.6% (95% CI: 4.8–7.0) of adolescents aged 12–18 years globally [2]. Although clinical diagnosis typically occurs during school years, accumulating evidence indicates that 70–80% of preschool children exhibiting ADHD symptoms maintain clinically relevant symptomatology into middle childhood [3, 4], highlighting the importance of early identification. However, the inherent behavioral variability characteristic of typical preschool development complicates reliable diagnosis at this critical stage [5], underscoring the need for objective behavioral markers to facilitate early screening.

Physical activity (PA) represents a promising behavioral domain for both early identification and intervention of ADHD [6, 7]. As a quantifiable manifestation of behavior, PA may reflect core ADHD-related characteristics, including executive dysfunction. Indeed, existing research has identified distinct PA profiles in children with ADHD, including greater overall PA volume [8], increased movement variability [9], higher activity levels [10]and reduced sedentary time [11] compared to typically developing peers. However, current evidence regarding PA patterns in ADHD presents apparent paradox. While some studies report elevated activity levels, others demonstrate reduced PA engagement or no significant differences. A critical synthesis of these studies suggests that these discrepancies are not arbitrary but may be attributed to several key factors. First, methodological heterogeneity plays a crucial role. Observational studies often rely on parent or teacher reports, which tend to capture macro-level, perceptible hyperactivity and may be biased by recall or situational specificity, frequently indicating reduced structured PA participation [12, 13]. In contrast, device-based accelerometry objectively quantifies micro-level movement patterns across the entire intensity spectrum, often revealing greater total volume and more variable movement in children with ADHD [8]. Second, environmental context exerts a regulatory influence. Structured environments (e.g., kindergarten settings) with adult supervision may suppress hyperactive behaviors compared to less-controlled environments (e.g., home settings), leading to context-dependent PA profiles. Third, and critically, the developmental stage of the sample is a likely pivotal variable. The majority of existing evidence is derived from school-aged children with established ADHD diagnoses. The preschool period represents a distinct neurodevelopmental stage characterized by rapid maturation of motor control and executive functions. The expression of ADHD-related motor signatures in this early period may differ qualitatively and quantitatively from later childhood of symptoms [14], yet it remains systematically understudied.

Consequently, the central question evolves from merely whether PA differs in ADHD to under what conditions, how, and at which developmental stage these differences emerge. The present study addresses this question directly by focusing on the understudied preschool population, employing objective accelerometry, and observing within the semi-structured kindergarten environment. This design allows us to characterize early, context-specific PA profiles associated with ADHD symptoms, thereby contributing data from a critical developmental window to help resolve the broader scientific debate.

Methods

Study design

This study is part of the prospective cohort of physical exercise intervention for preschoolers with ADHD symptoms. The clinical trial was registered with http://www.chictr.org.cn, ChiCTR2400083571.

Participants

From March to June 2025, a total of 84 children (18 with ADHD symptoms and 66 typical developing controls) from a public kindergarten in Shanghai were enrolled in this study. Eligible children with ADHD symptoms met the following criteria: 1) elevated ADHD symptoms (Factor IV or Factor VI scores $\geq \overline{x}$+1SD) on Conners Parental Symptom Questionnaire; 2) ADHD symptoms confirmed by a qualified and experienced pediatrician using DSM-5 diagnostic criteria; 3) aged between 3 and 6 years at recruitment, and 4) parents or legal guardians were willing to participate. Exclusion criteria included: 1) meeting criteria for autism spectrum disorder; 2) mental illness (primary mental illness, such as schizophrenia and bipolar disorder), 3) neurological disorders. In addition, a total of 66 neurotypical children were recruited from the same kindergarten. The inclusion criteria of the neurotypical group were age and sex matched with the ADHD group, but without any mental, neurological, or metabolic disease.

Measures

Conners parental symptom questionnaire

The Conners Parental Symptom Questionnaire (CPSQ) is a common tool used for screening ADHD symptoms. There are 48 items in the questionnaire, and the four-level scoring method (0, 1, 2, 3) could be summarized into six factors, including conduct problem (Factor I), learning problem (Factor II), somatic complaints (Factor III), hyperactivity (Factor IV), anxiety (Factor V), and Conners Index of hyperactivity (CIH, Factor VI). CPSQ has been introduced in China since 1980 s and has good

reliability and validity [15]. Positive CPSQ was defined as either Factor IV or Factor VI scores > ($\overline{x}$+1SD) in this study.

Physical activity

Physical activity was measured using the Actigraph w3GTX-BT accelerometer (ActiGraph Corp, Pensacola, FL, USA). The devices were secured on the right hip at 8:30 AM upon children's arrival at kindergarten and maintained continuous wear until 4:30 PM retrieval for five consecutive school days. Raw accelerometer data were collected at 30 Hz with 10 s epoch intervals [16]. Using Actilife (version 6.13.3) software to download and process raw accelerometer data, the Choi (2011) algorithm was employed to identify periods when children were not wearing the accelerometer [17]. Valid data is defined as a single day of valid wearing time ≥ 180 min and valid data days ≥ 3 days, which were included for further analysis. The Pate cutoff rule is used to classify physical activity as time spent in sedentary (0–799 counts per minute), light PA (800–1679 counts per minute), moderate PA (1680–3367 counts per minute), vigorous PA(≥ 3368 counts per minute) and moderate to vigorous PA (MVPA, ≥ 1680 counts per minute) for preschoolers.

Data collection

We used online surveys as methodology to collect demographic and questionnaire data via the WeChat network. Demographic data including age, sex, the only child, main caregiver, parental education level, and parental occupation were collected by self-administered questionnaire. Additionally, information on outside of kindergarten physical exercise (including type, frequency per week, and duration of each session) was simultaneously collected. Regular extracurricular physical exercise (EPE) was defined as structured sports activities occurring: 1) outside of kindergarten hours; 2) more than once a week; and 3) ≥ 30 min per session.

Ethical approval

The study was conducted according to the guidelines of the Declaration of Helsinki and approved by the institutional ethics committee of Minhang Hospital, Fudan University (2025–030-01 K).

Procedure

We used a two-step screening protocol to identify the early ADHD symptoms among preschoolers. Conners Parental Symptom Questionnaire was used for preliminary screening, where children who test positive undergo clinical evaluation. Clinical interviews were conducted in kindergarten by an experienced pediatric team, including interviews with parents, teachers, and patients. ADHD symptoms were assessed based on DSM-5 criteria. Formal diagnosis was not required given the well-documented challenges in establishing clinical ADHD diagnoses in preschool-aged populations [18, 19]. The flow diagram illustrating participant enrollment and assessment at each study phase is presented in Fig. 1.

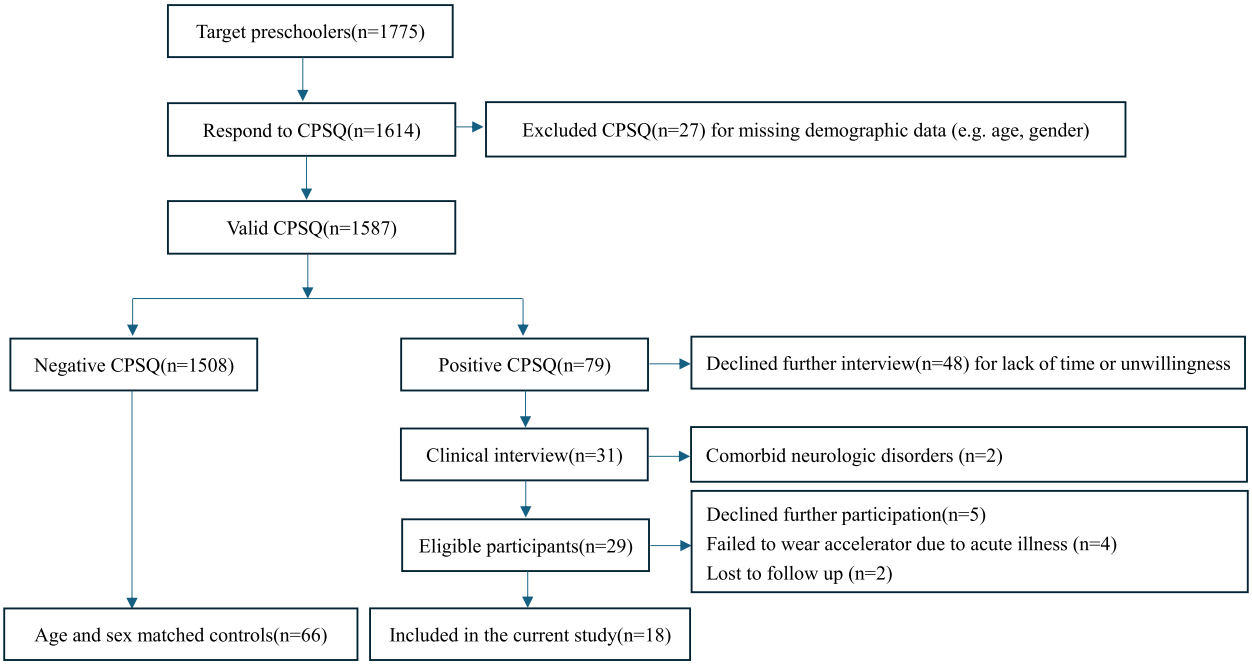


Fig. 1 Screening and identification flow chart of children with ADHD symptoms. Abbreviations: CPSQ, Conners Parental Symptom Questionnaire

Table 1 Basic characteristics of participants

Variables	Control group $n=66$	Cases group $n=18$	t/χ^2	P^1
Age (y)	5.39 ± 0.86	5.57 ± 0.98	−0.738	0.462
Gender (M: F)	38:28	12:6	0.485	0.486
The only child (yes, %)	43.94	77.78	-	0.016
The main caregiver				
Parents	46(69.70)	9(50.00)	-	0.069
Grandparents	20(30.30)	8(44.44)		
others	0	1(5.56)		
Mother's occupation:				
Technician	29(43.94)	9(50.00)	-	0.423
Freelancer	4(6.06)	2(11.11)		
Civil servant	7(10.61)	3(16.67)		
Other	26(39.39)	4(22.22)		
Father's occupation:				
Technician	24(36.36)	9(50.00)	-	0.837
Freelancer	10(15.15)	2(11.11)		
Civil servant	13(19.70)	3(16.67)		
Other	19(28.79)	4(22.22)		
Mother's education				
Undergraduate or below	13(19.70)	6(33.33)	3.786	0.151
Bachelor	39(59.09)	6(33.33)		
Postgraduate or above	14(21.21)	6(33.33)		
Father's education				
Undergraduate or below	7(10.61)	3(16.67)	-	0.749
Bachelor	37(56.06)	10(55.56)		
Postgraduate or above	22(33.33)	5(27.78)		
EPE (yes, %)	27(40.91)	11(61.11)	2.330	0.127
CPRS				
Factor IV	0.39 ± 0.37	1.56 ± 0.69	−9.573	< 0.001
Factor VI	0.36 ± 0.32	1.31 ± 0.35	−10.99	< 0.001

Abbreviations: ADHD attention-deficit hyperactivity disorder, *CPRS* Conners' Paret Rating Scale

[1]Performed by 2-sample t-test for normally distributed variables and chi-square test or Fisher exact test for categorical variables

Statistical analysis

STATA (Version 18.0, Stata Corp L.P.) was used for statistical analysis. Descriptive statistics were performed in total and by group. The characteristics of the study participants, including physical activity, are presented as the means (SDs) for normally distributed variables, medians (IQRs) for skewed variables and frequencies with percentages for categorical variables. Group comparisons were performed by 2-sample t tests, Two-sample Wilcoxon rank-sum (Mann–Whitney) tests and chi-square tests. Linear regression models were performed, with physical activity showing significant group differences ($P<0.05$) as the outcome variable and group (control group compared with the case group) as the exposure variable. To reduce confounding bias, multivariable linear regression models were performed to estimate coefficients with 95% confidence intervals (CIs) adjusted for age (y), gender, the only child status, main caregiver, parental education level (postgraduate or above, bachelor, undergraduate or below), parental occupation (Technician, Freelancer, Civil Servant, Other), and EPE (yes, no). To validate the in-kindergarten PA patterns, an interaction analysis was performed between EPE and group (EPE × group), controlling for potential confounders. Estimated margins with 95% CIs and Wald tests comparing groups stratified by EPE (yes or no) are reported.

Table 2 Characteristics of physical activity between control group and ADHD symptom group

Variables	Control $n=66$	Case $n=18$	Z	P^1
Average PA time (min)				
Sedentary	272.1(249.1, 298.6)	272.3(241.6, 297.1)	0.643	0.520
Light	27.3(23.2, 31)	24.1(21.3, 27)	2.268	0.023
Moderate	20.4(17.3, 24.2)	16. 6(13.9, 21)	2.595	0.009
Vigorous	8.1(6.7, 12.1)	6.6(5.6, 9.9)	1.695	0.090
MVPA	29.2(24.1, 36.8)	23.3(19.8, 33.8)	2.371	0.018

Abbreviations: ADHD attention-deficit hyperactivity disorder, *PA* physical activity, *MVPA* moderate to vigorous physical activity

[1]Two-sample Wilcoxon rank-sum (Mann–Whitney) test

Results

A total of 84 children (18 with ADHD symptoms and 66 without) with a mean age of 5.43 ± 0.89 years were included. Table 1 presents the baseline characteristics of the study participants. As shown, children in both groups were well-matched in terms of gender and age distribution. Other sociodemographic characteristics were generally balanced between groups, with no statistically significant differences observed.

Physical activity profiles

Children with ADHD symptoms exhibited significantly lower in-kindergarten physical activity levels compared to the control group (Table 2), particularly in light physical activity (LPA, 24.1 min; 25th, 75th: 21.3, 27 compared to 27.3; 25th, 75th: 23.2, 31, $P=0.023$), moderate physical activity (MPA, 16.6 min; 25th, 75th:13.9, 21 compared with 20.4 min; 25th, 75th: 17.3, 24.2, $P=0.009$), and moderate-to-vigorous physical activity (MVPA, 23.3 min; 25th, 75th:19.8, 33.8 compared to 29.2 min; 25th, 75th: 24.1, 36.8, $P=0.018$).

Results from multivariable regression models showed similar trends (Table 3).

Interactive associations with EPE

Analysis of predictive margins across groups and EPE revealed differing patterns of LPA, MPA, and MVPA by subgroup (Fig. 2). However, the formal interaction term

Table 3 Adjusted estimates of PA and interactive effects with EPE (yes, no)

	Without interaction		With interaction		Predictive margins (95%CI)			
	Mean difference (95% CI)		Mean difference (95% CI)					
	Group(1) *vs.* Group(0)		Group(1) *vs.* Group(0)		Group 0 ($n=66$)		Group 1 ($n=18$)	
	Adjusted[1]	P>|t|	Adjusted[2]	P>|t|	EPE (no)	EPE(yes)	EPE(no)	EPE(yes)
Light	−3.83(−7.62, −0.03)	0.048	−6.38(−10.17, −2.58)	0.001	27.89(25.85, 29.93)	27.23(25.03, 29.43)	21.51(18.51, 24.51)	25.57(20.64, 30.51)
Moderate	−4.76(−8.00, −1.52)	0.005	−6.51(−9.54, −3.49)	< 0.001	21.57(19.39, 23.75)	20.90(18.69, 23.12)	15.06(13.03, 17.08)	17.64(13.51, 21.76)
MVPA	−7.17(−11.74, −2.59)	0.003	−9.57(−13.52, −5.61)	< 0.001	31.30(28.09, 34.51)	30.40(27.12, 33.69)	21.73(19.03, 24.44)	25.28(18.92, 31.64)

Abbreviations: PA physical activity, *EPE* extracurricular physical exercise; typically developing; Group1 = with ADHD symptom group; Group = 0 = control group

[1]Estimated from multivariable regression without an interaction term, adjusted for age (years), gender, the only child(yes, no), main caregiver, extracurricular physical exercise(EPE; yes, no), and the first two principal components derived from multiple correspondence analysis of parental educational attainment (Postgraduate or above, Bachelor, undergraduate or below) and parental occupational categories (Technician, Freelancer, Civil Servant, Other)

[2]The results were estimated via multivariable regression with an interaction term between group and extracurricular physical activity, adjusted for age (years), gender, the only child(yes, no), main caregiver, extracurricular physical exercise(EPE; yes, no), and the first two principal components derived from multiple correspondence analysis of parental educational attainment (Postgraduate or above, Bachelor, undergraduate or below) and parental occupational categories (Technician, Freelancer, Civil Servant, Other)

(Group × EPE) in the model was not statistically significant ($P>0.05$ for all outcomes). Wald's tests confirmed significant main effects of group on LPA ($F=11.24$, $P=0.001$), MPA ($F=18.45$, $P<0.001$) and MVPA ($F=23.21$, $P<0.001$).

Discussion

To our knowledge, this study provides the first objective evidence characterizing PA patterns in preschool children with ADHD symptoms. Our accelerometer data suggests that preschoolers with ADHD symptoms may exhibit significantly lower engagement in light (LPA), moderate (MPA), and moderate-to-vigorous physical activity (MVPA).

This aligns with studies reporting reduced PA participation in older ADHD populations [13, 20], suggesting that PA deficits may emerge early in development.

Notably, our results contrast with studies reporting higher overall PA volume or reduced sedentary time in school-aged children with ADHD [9–11]. This discrepancy may be explained by these factors highlighted in our introduction. First, methodological differences in accelerometer use may play a role. While our study focused on kindergarten hours, previous 24-h/7-day monitoring studies captured total daily movement [8], potentially masking context-dependent PA variations. Second, environmental influences are especially important in early childhood. Preschool settings typically enforce stricter behavioral rules than home environments. This may suppress spontaneous movement. Finally, developmental differences are key. Older children with ADHD often express hyperactivity through frequent movement [21]. In contrast, preschoolers with ADHD symptoms may lack sufficient motor skills or behavioral control for consistent PA participation. These findings underscore the importance of age-specific PA interventions tailored to early ADHD symptoms.

Our findings extend previous work by revealing PA reductions are detectable even in preschool ADHD symptoms. ADHD children tend to engage in less PA [22], which may be associated with executive dysfunction (e.g., impaired planning and inhibitory control) [23, 24]. Some evidence suggests a potential bidirectional relationship, where executive function deficits could limit PA participation, while low PA exacerbates EF impairments (e.g., Tower of London task performance: adjusted $R^2=0.28$, $P<0.05$) [25]. Besides, ADHD frequently co-occurs with comorbidities that compound PA reduction. Insufficient PA mediates the longitudinal link between childhood ADHD symptoms and adolescent obesity (OR = 1.91, 95%CI: 1.10–3.33) [26]. These findings raise the possibility of a cyclical pattern: ADHD-reduced PA-weight gain-further activity decline [27]. Also, concurrent emotional dysregulation (e.g., anxiety/

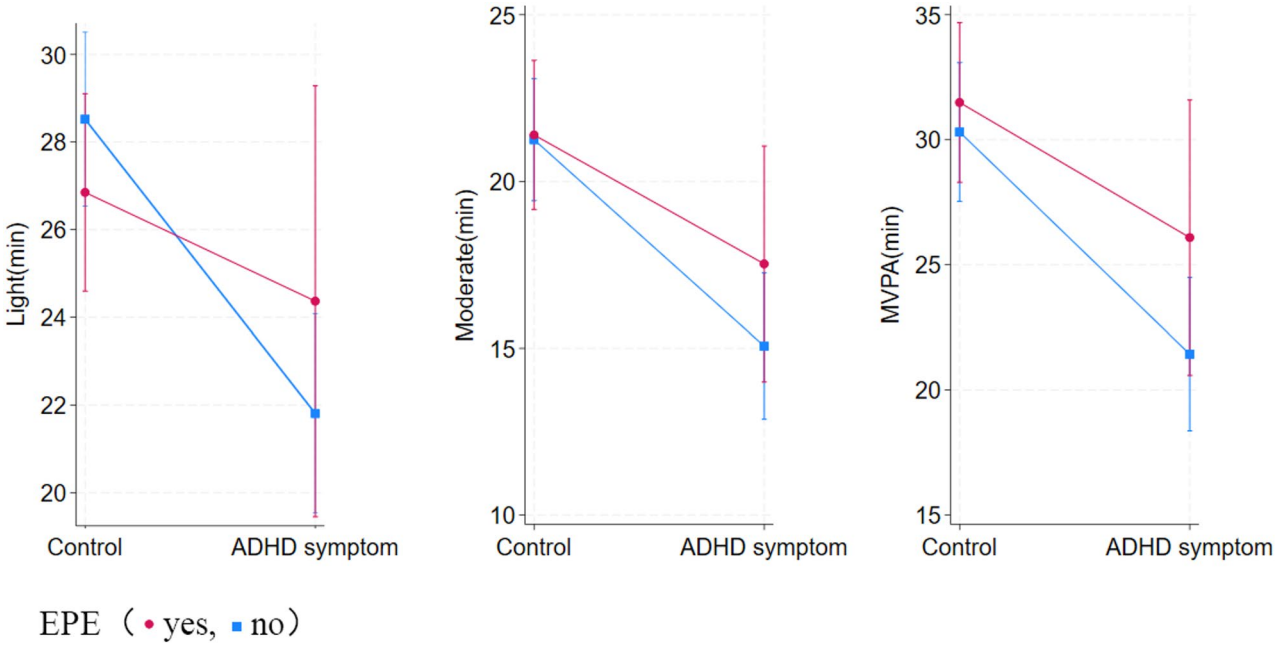


Fig. 2 Plots of estimated margins between Control and ADHD symptom groups and their interaction with EPE. ADHD, attention-deficit hyperactivity disorder; EPE, extracurricular physical exercise

depression) can reduce an individual's motivation and energy, thereby decreasing the possibility of participating in physical activities [28, 29]. From a neurobiological perspective, these patterns could be linked to dopamine dysfunction [30]. The dopamine system plays key roles in cognitive control and motivation. In ADHD, this system's dysregulation may lead to reduced motivation for effortful activities. Specifically, individuals with ADHD may show decreased engagement in exercise, particularly when rewards are delayed [31]. Supporting this, MAOA genotype variations influence ventral striatum activation patterns (reward circuitry) [32], suggesting genetic modulation of activity-related behaviors.

Extracurricular physical exercise (EPE) and in-kindergarten physical activities are both important components of movement behaviors in preschool children [33]. In our study, we found no significant interaction between EPE and group differences in kindergarten PA levels ($P > 0.05$). This suggests that lower PA levels in preschoolers with ADHD symptoms were not significantly linked to their overall daily activity patterns. The reduced PA likely reflects kindergarten-specific challenges, including structured participation deficits [20] and impaired social interactions [34]. These findings support tailored educational strategies to promote PA independent of their baseline activity levels.

This study has several limitations that should be acknowledged. First, the most notable limitation is the modest sample size, particularly of the group with ADHD symptoms ($n = 18$), which necessitates cautious interpretation of the findings. This sample size, however, reflects two fundamental challenges in preschool ADHD research:1) The epidemiological context: The population prevalence of clinically significant ADHD symptoms in preschoolers is estimated to be low (approximately 1.9–3.3%) [35, 36]. 2) The rigor of ascertainment: Our two-step screening procedure was employed to enhance diagnostic precision and reduce false positives, a critical concern in preschool populations where normative high activity overlaps with pathology. The main implication of this sample size is reduced statistical power, increasing the risk of unstable estimates. Nevertheless, confidence in the primary between-group findings is bolstered by key methodological strengths: the use of objective, high-resolution accelerometry; the well-characterized control group drawn from the same environment; and the consistency of the observed effect directions with larger-scale studies in older children with ADHD. Therefore, while the findings regarding global PA reductions in children with ADHD symptoms appear reliable within this context, they should be viewed as robust preliminary evidence that necessitates replication in larger, multi-site cohorts. Second, the participants were recruited only from public kindergartens in Shanghai, potentially limiting the representativeness of the results for other preschool-aged children, particularly those from diverse socioeconomic or cultural backgrounds. Third, while accelerometers provide objective measures of physical activity, they cannot capture contextual factors (e.g., structured vs. unstructured activities) that may influence movement patterns in children with ADHD symptoms. Fourth, some evidence indicates that symptoms

of inattentiveness, rather than hyperactivity-impulsivity symptoms, tend to correlate with reduced physical activity levels [9, 37]. Our study did not conduct subtype classification of preschool ADHD symptoms. We could not assess whether specific symptom domains differentially relate to PA engagement. Future investigations with larger samples and dimensional symptom assessments are warranted. Fifth, the cross-sectional design precludes causal inference regarding the direction of the observed relationship. Future multi-center longitudinal studies with larger, diverse samples are essential to establish causality and to examine whether reduced PA is a precursor, correlate, or consequence of emerging ADHD symptomatology in early childhood.

In conclusion, this study reveals that preschool children with ADHD symptoms exhibit significantly lower PA. These findings highlight the importance of early identification and targeted PA interventions to promote their overall health. Future research should further clarify the causal relationship between these factors and explore potential intervention strategies in this population.

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Clinical trial
Clinical trial number: ChiCTR2400083571(registered on 28 April 2024 at http://www.chictr.org.cn/).

Authors' contributions
Xiaofeng Yin and Yanqi Hu designed the study. Xueqing Miao, Mingyuan Fan, Xiyuan Zeng and Wangsong Li contributed to the interpretation of data. Lili Zhang wrote and revised the main manuscript text. Cun You and Jianying Ma carried out the statistical analysis and drafted the manuscript. Xueqing Miao and Mingyuan Fan collected the data. Xiaofeng Yin and Yanqi Hu supervised and revised the manuscript. All authors reviewed the manuscript. The authors read and approved the final manuscript.

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Data availability
The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
This study was approved by the institutional ethics committee of Minhang Hospital, Fudan University (2025–030-01 K) and adhere to the Helsinki Declaration in the course of implementation. All children and their parents gave written informed consent.

Consent for publication
Not applicable.

Competing interests
The authors declare no potential conflicts of interest with respect to the research, authorship, or publication of this article.

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